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Q Find: $L\{f(t)\}$
 where $f(t) = \begin{cases} 0 & 0 \leq t \leq 3 \\ 2 & t \geq 3 \end{cases}$

Sol: $L\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt$

$$= \int_0^3 e^{-st} f(t) dt + \int_3^{\infty} e^{-st} f(t) dt$$

$$= \int_0^3 e^{-st} \times 0 dt + \int_3^{\infty} e^{-st} \times 2 dt$$

$$= 2 \int_3^{\infty} e^{-st} dt$$

$$= 2 \left[\frac{e^{-st}}{-s} \right]_3^{\infty}$$

$$= 2 \left[0 - \left(\frac{e^{-3s}}{-s} \right) \right] = \frac{2}{s} e^{-3s}$$

$$s > 0.$$

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Inverse Laplace Transform:

If $L\{f(t)\} = f(s)$ Then $f(t)$
 is called inverse laplace
 transform of $f(s)$.

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ie... $f(t) = L^{-1} \{F(s)\}$

$F(s) = L^{-1} \{F(s)\} = f(t)$

- $1/s \Rightarrow 1$
- $\frac{n!}{s^{n+1}} \Rightarrow t^n$ • $\frac{1}{s^2} \Rightarrow t$
- $\frac{k}{s^2 + k^2} \Rightarrow \sin kt$
- $\frac{s}{s^2 + k^2} \Rightarrow \cos kt$
- $\frac{1}{s - k} \Rightarrow e^{kt}$
- $\frac{k}{s^2 - k^2} \Rightarrow \sinh kt$
- $\frac{s}{s^2 - k^2} \Rightarrow \cosh kt$

★ First shifting theorem for inverse

$$L^{-1} \{F(s-k)\} = e^{kt} f(t) = e^k L^{-1} \{f(s)\}$$

Q show that

$$L^{-1} \left\{ \frac{s+1}{s(s^2+1)} \right\} = 1 + \sin t - \cos t$$

$$\frac{A}{s} + \frac{Bx+C}{(s^2+1)} + \frac{Dx+E}{(s^2+1)^2}$$

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Sol: By Partial fraction.

$$\frac{s+1}{s(s^2+1)} = \frac{1}{s} + \frac{1}{s^2+1} - \frac{s}{s^2+1}$$

$$L^{-1} \left\{ \frac{s+1}{s(s^2+1)} \right\} = L^{-1} \left\{ \frac{1}{s} \right\} + L^{-1} \left\{ \frac{1}{s^2+1} \right\}$$

$$- L^{-1} \left\{ \frac{s}{s^2+1} \right\}$$

$$= 1 + \sin t - \cos t.$$

Prove that:

$$L^{-1} \left\{ \frac{s+1}{s^2(s^2+s+1)} \right\} = t - 2 \frac{e^{-t/2}}{\sqrt{3}} \sin \sqrt{3}/2 t$$

Sol: By Partial fractional

$$\frac{s+1}{s^2(s^2+s+1)} = \frac{1}{s^2} - \frac{1}{s^2+s+1}$$

$$L^{-1} \left\{ \frac{s+1}{s^2(s^2+s+1)} \right\} =$$

$$L^{-1} \left\{ \frac{1}{s^2} \right\} - L^{-1} \left\{ \frac{1}{s^2+s+1} \right\}$$

$$= t - L^{-1} \left\{ \frac{1}{(s^2+s+1/4) + (\sqrt{3}/2)^2} \right\}$$

$$1 - \frac{1}{4} = \frac{3}{4} \left(\frac{\sqrt{3}}{2} \right)^2 \quad 1 \rightarrow \frac{1}{4} + \left(\frac{\sqrt{3}}{2} \right)^2$$

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$$= t - \frac{2}{3} L^{-1} \left\{ \frac{\sqrt{3}/2}{(s+1/2)^2 + (\sqrt{3}/2)^2} \right\}$$

$$= t - \frac{2}{\sqrt{3}} e^{-t/2} \sin \frac{\sqrt{3}}{2} t.$$

$$\therefore \sin kt = \frac{k}{s^2 + k^2}$$

$$\therefore \frac{\sin \sqrt{3} t}{2} = \frac{\sqrt{3}/2}{s^2 + \left(\frac{\sqrt{3}}{2} \right)^2}$$

Show that

$$L^{-1} \left\{ \frac{s^4 - 2s^3 + 2}{s^3(s^2 - s - 2)} \right\}$$

$$= -\frac{3}{4} + \frac{t}{2} - \frac{t^2}{2} + \frac{1}{3} e^{-t} + \frac{17}{12} e^{2t}$$

Find the inverse Laplace transform of

$$F(s) = \frac{s^4 + 2s^3 + 2}{s^3(s^2 - s - 2)}$$

→ By partial fractional

$$= -\frac{3}{4} \cdot \frac{1}{s} + \frac{1}{2} \cdot \frac{1}{s^2} - \frac{1}{s^3} + \frac{1}{3} \cdot \frac{1}{s+1} +$$

$$\frac{17}{12} \cdot \frac{1}{s-2}$$

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$$s^3(s^2 - s - 2) = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{s-2}$$

$$= \frac{A}{s} + \frac{B}{s^2 + s + \frac{C}{s^2}} + \frac{C}{(s-1)^2} + \frac{C}{(s-2)^2}$$

$$= -\frac{3}{4} L^{-1} \left\{ \frac{1}{s} \right\} + \frac{1}{3} L^{-1} \left\{ \frac{1}{s+1} \right\} + \frac{17}{12}$$

$$L^{-1} \left\{ \frac{1}{s^3} \right\} + \frac{1}{3} L^{-1} \left\{ \frac{1}{s+1} \right\} + \frac{17}{12}$$

$$L^{-1} \left\{ \frac{1}{s-2} \right\}$$

$$= \frac{3}{4} + \frac{t}{2} - \frac{t^2}{2} + \frac{1}{3} e^{-t} + \frac{17}{12} e^{2t}$$

Q Find $L^{-1} \left\{ \frac{1}{(s-5)^3} \right\} = e^{5t} \cdot L^{-1} \left\{ \frac{1}{s^3} \right\}$

show that

X By partial fractional.

$$\frac{A}{(s-5)} + \frac{B}{(s-5)^2} + \frac{C}{(s-5)^3} \times$$

$$= \frac{1}{2} e^{5t} L^{-1} \left\{ \frac{2}{s^3} \right\}$$

$$= \frac{1}{2} e^{5t} t^2$$

Q Find $L^{-1} \left\{ \frac{1}{s^2 + 2s} \right\}$

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$$\Rightarrow L^{-1} \left\{ \frac{1}{s^2 + 2s} \right\}$$

$$= L^{-1} \left\{ \frac{1}{s^2 + 2s + 1 - 1} \right\}$$

$$= L^{-1} \left\{ \frac{1}{(s+1)^2 - (1)^2} \right\}$$

$$= e^{-t} \sin ht.$$

$$\underline{\text{Q:}} \quad L^{-1} \left\{ \frac{1}{(s-a)(s-b)} \right\} \quad \because \begin{matrix} s^2 - sb - \\ sa + ab \end{matrix}$$

By partial fraction

$$L^{-1} \left\{ \frac{1}{(s-a)(s-b)} \right\} = \frac{A}{s-a} + \frac{B}{s-b}$$

$$1 = A(s-b) + B(s-a)$$

$$s-b=0$$

$$s=b$$

$$s-a=0$$

$$s=a$$

$$1 = B(b-a)$$

$$B = \frac{1}{(b-a)}$$

$$A = \frac{1}{a-b}$$

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$$L^{-1} \left\{ \frac{1}{(s-a)(s-b)} \right\} = \frac{1}{(a-b)(s-a)} + \frac{1}{(s-b)(b-a)}$$

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$$= \frac{1}{a-b} \left\{ \frac{1}{(s-a)} - \frac{1}{(s-b)} \right\}$$

$$\mathcal{L}^{-1} \left\{ \frac{1}{(s-a)(s-b)} \right\} = \frac{1}{a-b} \mathcal{L}^{-1} \left\{ \frac{1}{s-a} - \frac{1}{s-b} \right\}$$

$$= \frac{1}{a-b} (e^{at} - e^{bt}).$$